



GUI Development in Java

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Introduction to GUI in Java

GUI stands for **Graphical User Interface**

Allows users to interact using:

- Buttons
- Text fields
- Windows

Alternative to command-line (console) interaction

GUI Libraries in Java

1. AWT (Abstract Window Toolkit)

- Oldest GUI library
- Uses OS-based components

2. Swing

- Built on AWT
- More powerful and flexible
- Platform-independent

AWT Overview

Provides basic GUI components:

- Button
- Label
- TextField
- Frame

Limitation:

- Depends on operating system
- Limited customization

Swing Overview

- ▶ Modern GUI library in Java
- ▶ Rich set of components
- ▶ **Common Components:**
 - ▶ JFrame → window
 - ▶ JButton → button
 - ▶ JLabel → text display
 - ▶ JTextField → input field
- ▶ **Advantage:** Consistent look across platforms



Basic Steps to Create GUI



Create a window (JFrame)



Add components (button, label, etc.)



Set size and visibility



Handle user events

Simple GUI Example

```
import javax.swing.*;

public class SimpleGUI {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Java GUI Example");
        JButton button = new JButton("Click Me");

        frame.add(button);
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```



Event Handling



GUI works on **event-driven programming**



Actions happen when user interacts



Examples:

Button click
Mouse click
Key press

Common GUI Components

JButton → perform action

JLabel → display text

TextField → user input

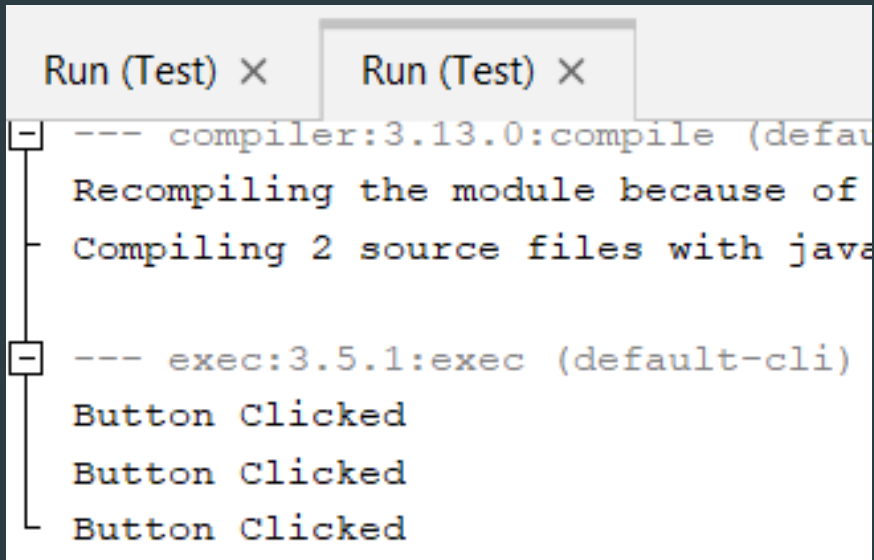
TextArea → multi-line input

JCheckBox → multiple selection

JRadioButton → single selection

Button Click Event Monitoring

- ▶ When the button is clicked → message prints in console:



The screenshot shows an IDE console window with two tabs labeled "Run (Test) X". The console output is as follows:

```
--- compiler:3.13.0:compile (default-compile)
Recompiling the module because of
Compiling 2 source files with java
--- exec:3.5.1:exec (default-cli)
Button Clicked
Button Clicked
Button Clicked
```

```
import javax.swing.*;
import java.awt.*;

public class Test {
    public static void main(String[] args) {

        // Create frame (window)
        JFrame frame = new JFrame("Button Example");

        // Create button
        JButton button = new JButton("Click Me");

        // Add action listener (event handling)
        button.addActionListener(e -> {
            System.out.println("Button Clicked");
        });

        // Set layout and add button
        frame.setLayout(new FlowLayout());
        frame.add(button);

        // Set frame properties
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

javax Package in Java



javax stands for **Java Extension Packages**



Provides additional libraries beyond core java packages



javax.swing → GUI components (JFrame, JButton, JLabel)



javax.servlet → Web applications (Servlets)



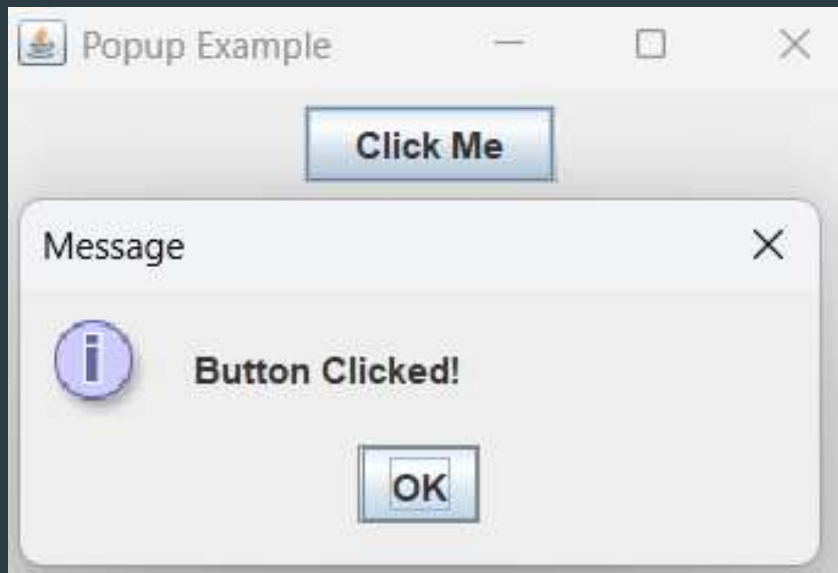
javax.xml → XML processing



javax.persistence → Database (JPA)

Button Click Event Monitoring on GUI

- ▶ When clicked → shows a popup message: "Button Clicked!"



```
import javax.swing.*;
import java.awt.*;

public class Test {
    public static void main(String[] args) {

        // Create frame
        JFrame frame = new JFrame("Popup Example");

        // Create button
        JButton button = new JButton("Click Me");

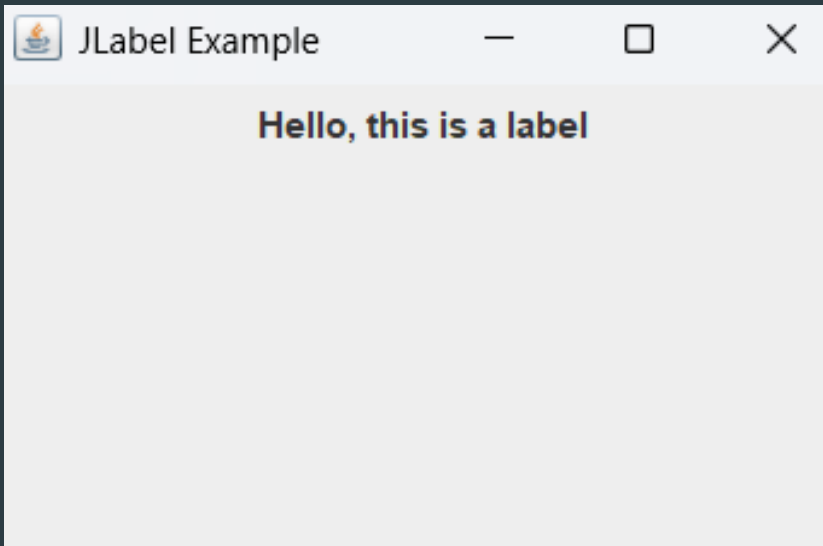
        // Add action listener (event handling)
        button.addActionListener(e -> {
            JOptionPane.showMessageDialog(frame, "Button Clicked!");
        });

        // Set layout and add button
        frame.setLayout(new FlowLayout());
        frame.add(button);

        // Frame settings
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

JLabel (Text Display)

- Displays simple text inside the window.



```
import javax.swing.*;  
import java.awt.*;
```

```
public class Test {  
    public static void main(String[] args) {  
  
        JFrame frame = new JFrame("JLabel Example");  
  
        // Create label  
        JLabel label = new JLabel("Hello, this is a label");  
  
        // Layout and add  
        frame.setLayout(new FlowLayout());  
        frame.add(label);  
  
        // Frame settings  
        frame.setSize(300, 200);  
  
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
        frame.setVisible(true);  
    }  
}
```

JTextField (User Input)

- Takes input from user and shows it in a popup.



```
import javax.swing.*;  
import java.awt.*;
```

```
public class Test{  
    public static void main(String[] args) {  
  
        JFrame frame = new JFrame("TextField Example");  
  
        JTextField textField = new JTextField(15);  
        JButton button = new JButton("Show Text");  
  
        // Event handling  
        button.addActionListener(e -> {  
            String input = textField.getText();  
            JOptionPane.showMessageDialog(frame, "You  
entered: " + input);  
        });  
  
        frame.setLayout(new FlowLayout());  
        frame.add(textField);  
        frame.add(button);  
  
        frame.setSize(300, 200);  
  
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
        frame.setVisible(true);  
    }  
}
```

JRadioButton (Single Selection)

- Only one option can be selected at a time.



```
import javax.swing.*;
import java.awt.*;

public class CheckBoxExample {
    public static void main(String[] args) {

        JFrame frame = new JFrame("CheckBox Example");

        JCheckBox cb1 = new JCheckBox("Java");
        JCheckBox cb2 = new JCheckBox("Python");
        JButton button = new JButton("Submit");

        button.addActionListener(e -> {
            String result = "Selected: ";
            if (cb1.isSelected()) result += "Java ";
            if (cb2.isSelected()) result += "Python";

            JOptionPane.showMessageDialog(frame, result);
        });

        frame.setLayout(new FlowLayout());
        frame.add(cb1);
        frame.add(cb2);
        frame.add(button);

        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

Application: Login Form

- JLabel → shows text
- JTextField → username input
- JPasswordField → password input
- JButton → performs action
- JOptionPane → shows result



```
import javax.swing.*;
import java.awt.*;

public class LoginForm {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Login Form");

        // Components
        JLabel userLabel = new JLabel("Username:");
        JTextField userField = new JTextField(15);

        JLabel passLabel = new JLabel("Password:");
        JPasswordField passField = new JPasswordField(15);

        JButton loginButton = new JButton("Login");

        // Action
        loginButton.addActionListener(e -> {
            String username = userField.getText();
            String password = new String(passField.getPassword());

            if (username.equals("admin") && password.equals("1234")) {
                JOptionPane.showMessageDialog(frame, "Login Successful!");
            } else {
                JOptionPane.showMessageDialog(frame, "Invalid Credentials!");
            }
        });

        // Layout
        frame.setLayout(new FlowLayout());

        frame.add(userLabel);
        frame.add(userField);
        frame.add(passLabel);
        frame.add(passField);
        frame.add(loginButton);

        // Frame settings
        frame.setSize(300, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```